

Robots in Context

Programming Exercise 1
Topic: ROS

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Scope of the Exercise

The general aim

- Continue exploring ROS with Turtlesim as a sandbox
- Learn about Actions
- Familiarize you with the tools in ROS



Programming Exercise 1 - Tasks

- Read up on actions
<https://docs.ros.org/en/jazzy/Tutorials/Beginner-CLI-Tools/Understanding-ROS2-Actions/Understanding-ROS2-Actions.html>
- Explore the Turtlesim tutorial
<https://docs.ros.org/en/jazzy/Tutorials/Beginner-CLI-Tools/Introducing-Turtlesim/Introducing-Turtlesim.html>
 - Create new turtles using the service
 - Control turtle(s) via the teleop_turtle node
 - Inspecting data being published
 - Send goals
- Visualize data, e.g. position, in rqt
 - Also, inspect your setup using rqt_graph (incl. topics)
(note: rqt and rqt_graph are different tools)
- Write your own node that calculates the speed of your turtle
- Bonus: can you visualize your turtles un RVIZ? Why/why not or how?
- Optional: Explore the parameters



Programming Exercise 1 - Reporting

- General: 7ish minute presentation:
 - Prepare an explanatory walk through of actions
 - Provide an overview of your work
 - Consider, e.g., to take some screenshots when applicable
 - Avoid running through source code too much, short, interesting snippets might be valuable though
 - Mention also challenges
 - This can be valuable for others too and likely there will be others that will face the same problems you did
 - Share presentation material on ItsLearning
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Evaluation

There is no grading or pass/fail assessment. Consider the following (as a generic guideline):

1. Is the theoretical concept clearly conveyed?
2. Is goal of the activity well-defined?
3. Is development work explained in an undestandably?
(would fellow students be able to reproduce the work?)
4. If applicable, is the data sufficient to support an analysis?
5. Is the conclusion coherent? Are potential unexpected findings analysed / investigated?

